Band-1 Hadamard Order 6404 — Miyamoto (Kronecker Seed H_1600 + C_1602 over prime 1601); Miyamoto-only Bonus (NOT in SPK); Gap 2052 Still OPEN

H_20/H_400/H_1600 Kronecker PASS; C_1602/H_6404 Miyamoto PASS; classical SPK on 6404 FAIL; 6404 Miyamoto-only (NOT in SPK); 16×400=6400≠6404; gap 2052 still OPEN; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

2026-09-15

Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list · Human+AI attack order · ZoraASI mission-control.
Parents. `2026-09-15_Band1_Hadamard_6388_Miyamoto`; `2026-09-15_Band1_Hadamard_6212_Miyamoto`.
Lab. `/workspace/experiments/band1/hadamard_6404_kronecker/`.
Cutoff. 2026-09-15T23:47:40Z (~7:47PM ET).

Honesty stamp (non-negotiable).
This memo claims **finite constructions of orders 20, 400, 1600, and 6404 only** (verified max $|HH^\top - nI| = 0$), via Kronecker seed chain $H_{20} =$ Paley-I$(q = 19)$, $H_{400} = H_{20} \otimes H_{20}$, $H_{1600} = H_4 \otimes H_{400}$, and Miyamoto $H_{6404} =$ Miyamoto(H_{1600}, C_{1602}) with symmetric Paley conference C_{1602} over prime $q = 1601 \equiv 1 \pmod 4$.

Arithmetic correction (honesty). Prior next-target shorthand '`Kronecker$(16\times 400)$ / Kronecker$(4\times 1600)$''` constructs order **6400**, not 6404. Order `$6404=4\cdot 1601$`; sole classical factor pair `$(4,1601)$` with `$1601$` not Hadamard-admissible. Classical SPK paths for 6404 **FAIL**: Paley-I (`$q=6403=19\cdot 337$` not prime); Paley-II (`$q=3201=3\cdot 11\cdot 97$` not prime); Kronecker (`$4\times 1601$`). Therefore order **6404 is a Miyamoto-only bonus (NOT in SPK)**. Seed **1600** (and 400/20) **are** in SPK via Kronecker/Paley-I. Related order **6400** is in SPK via Kronecker but is **not** the 6404 target. This memo **does not** claim the full Hadamard conjecture. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P`$\neq$`NP / YM / Frankl / NS-solved / AC-solved / Hadamard-conjecture-solved claims. Standard constructive gap **2052**

remains **OPEN** under Sylvester$\cup$Paley$\cup$Kronecker (SPK) closure.clay_closed = false‘.

Abstract

Finite constructions of orders **20**, **400**, **1600**, and **6404**.

- H_{20} = Paley-I ($q = 19$)
- H_{400} = Kronecker(H_{20}, H_{20})
- H_{1600} = Kronecker(H_4, H_{400}) — **in SPK**
- C_{1602} = Paley conference over prime 1601
- H_{6404} = Miyamoto(H_{1600}, C_{1602})
- Classical Paley-I / Paley-II / Kronecker for 6404: **FAIL / SKIP**

SPK: 6404 Miyamoto-only bonus (NOT in SPK). Seed 1600 enters SPK via Kronecker. Related 6400 = Kronecker(H_{16}, H_{400}) is SPK but $\neq 6404$. Prior SPK seeds through 6388 ship (incl. 1596) retained; **6388/6404 NOT folded**.
Gap 2052: OPEN. **Conjecture:** OPEN. **clay_closed:** false. **GATE-04:** LOCKED.

`python3 verify_hadamard_6404.py`

Results. overall_pass=true; units 81/81; wall 214.1596s; all max|HH^T-nI|=0.

Cutoff: 2026-09-15T23:47:40Z (~7:47PM ET)

Arithmetic

- Target $n = 6404 = 4 \cdot 1601$; 1601 prime $\equiv 1 \pmod 4$ → Miyamoto-eligible.
- Need H_{1600} and C_{1602} (Paley conference over prime 1601).
- Seed: Kronecker $H_{1600} = H_4 \otimes H_{400}$ with $H_{400} = H_{20} \otimes H_{20}$, $H_{20} =$ Paley-I(19).
- Direct Paley-I/II for 1600: **FAIL** ($q = 1599$, $q = 799$ composite).
- SPK for 6404: Paley-I $q = 6403$ **FAIL**; Paley-II $q = 3201$ **FAIL**; Kronecker **FAIL** → **Miyamoto-only**.
- Correction: $16 \cdot 400 = 6400 \neq 6404$; $4 \cdot 1600 = 6400 \neq 6404$.

Constructions (verified)

Object	Method	max|HH^T-nI|	Wall (s)
H_20 Paley-I	Paley-I q=19	0	0.0002
H_400 Kronecker	Kronecker(H_20, H_20)	0	0.0357
H_1600 Kronecker	Kronecker(H_4, H_400)	0	2.0005

Object	Method	max|HH^T-nI|	Wall (s)
C_1602	Paley conf q=1601	0	2.8936
H_6404 Miyamoto	Miyamoto(H_1600, C_1602)	0	207.7946
H_6404 Paley-I	SKIP q=6403=19·337 not prime	—	—
H_6404 Paley-II	SKIP q=3201=3·11·97 not prime	—	—
H_6404 Kronecker	SKIP (4·1601; 1601 not Hadamard-admissible)	—	—
H_6400 related	Kronecker(16×400) arithmetic doc (≠6404)	—	—

SPK closure note

Seeds include prior SPK ships through **6388** lineage (**1596** from that ship; **NOT** 6388/6404) + verified **20/400/1600** (Kronecker/Paley-I); **6404 NOT folded** (Miyamoto-only); related **6400** SPK-admissible arithmetically; **exclude** Miyamoto-only bonuses including **6404/6388/6196/5956/5812/5476**. Known count 1314. Next constructive gap **2052** (OPEN).

Quantity	Value
6404 in SPK	**False** (Miyamoto-only bonus)
1600 in SPK	**True** (Kronecker 4×400)
400 in SPK	**True** (Kronecker 20×20)
6400 related in SPK	**True** (Kronecker 16×400; ≠6404)
1596 in SPK	**True** (prior Paley-II)
6388 in SPK	**False** (prior Miyamoto-only)
Next constructive gap	**2052** (OPEN)

Next Miyamoto targets (after q=1601)

q	n=4q	need H	build notes
1609	6436	1608	[‘PaleyI(q=1607)’]
1613	6452	1612	[]
1621	6484	1620	[‘PaleyI(q=1619)’, ‘PaleyII(q=809)’]
1637	6548	1636	[]
1657	6628	1656	[‘Double(PaleyI half=828)’]

Honesty

clay_closed = false · GATE-04 LOCKED · Hadamard conjecture OPEN · gap 2052 OPEN · no RH/P!=NP/YM/Frankl/NS/AC/Clay-closed claims · **no false claim that 6404 ∈ SPK via Kronecker**.